Battery Module Studio

Formulas, calculations and graph definitions

TELIPORT Season 4: Reusable Digital Platform for Designing Battery Energy System
Cell: LG INR21700-M50 — Simulation only — 31 October 2026

How to use this document. Every number the dashboard shows is produced by a formula written here, in the order the code evaluates it. Each section names the source file. Parameter sources are tagged: *DS* is the LG Chem product specification INR21700 M50 18.20Wh (2016-08-23, status "Tentative"), with section number; *BD* is the batterydesign.net page "LG INR 21700 M50" (5 May 2023), mainly its DCIR-versus-temperature chart; *PB* is the PyBaMM `Chen2020` parameter set (Chen et al., J. Electrochem. Soc. 167, 080534, 2020); *FIT* is a value fitted offline to PB; *assumption* is in none of these, and the reason is given. Numbers in the worked examples use the default settings: 4S3P module, 35 °C ambient, 1C discharge, 0.6C charge, two back-to-back cycles, 5 % resistance spread, seed 7.

What the sources do not contain. The M50 specification is a tentative draft. It has no open-circuit-voltage curve, no capacity-versus-rate table, no charge-time data and no thermal data. The platform therefore takes its OCV curve and its diffusion behaviour from PB, its resistance temperature dependence from a chart in BD, and says in Section 16 which validation rows are independent checks and which only reproduce an input.

Contents

1 Model structure **3**

2 Notation **3**

3 Cell data and their sources **4**

4 Electrical sub-model **4**

4.1 Open-circuit voltage 5
4.2 Resistance, temperature and cell spread 5
4.3 Fitting R_0, R_1, τ 6
4.4 Polarisation state 6
4.5 Parallel groups and current sharing 6
4.6 Rates and state of charge 6
4.7 Discharge 7
4.8 Constant-current, constant-voltage charge 7
4.9 Current limits as a check, not a clamp 7

5 Heat generation **7**

6 Thermal network **8**

6.1 Heat capacity 8

6.2 Convection to ambient . . . 8
6.3 Conduction between neighbouring cells . . . 9
6.4 Spreader plate (design B) . . . 9
6.5 Governing equation and solver . . . 9
6.6 Time constants and a hand check . . . 9

7 BMS and protection logic 10

8 Duty cycle and time stepping 10

9 Indicators 11

10 Mass, cost and volume 12

11 Checks 12

12 Electrochemical layer: lithium-plating screen 13

13 Liquid cooling: Design D 15

13.1 Nodes and heat paths . . . 16
13.2 Heat-exchanger conductances . . . 16
13.3 Governing equations . . . 16
13.4 Auxiliary energy, heat and net usable energy . . . 17
13.5 Checks added for the liquid loop . . . 17
13.6 Parameters and sources . . . 18
13.7 Result: four cooling systems, default case . . . 18
13.8 Ambient temperature, discharge rate and repeated cycles . . . 20
13.9 Fault cases . . . 21
13.10 Safety and engineering risks outside the model . . . 21
13.11 Limits of the liquid model . . . 22

14 Ranking and recommendation 22

15 Graphs and tables: what is plotted 23

15.1 Hottest cell temperature over the duty cycle . . . 23
15.2 Indicators against requirements . . . 24
15.3 Cell temperature map . . . 24
15.4 Electrical behaviour . . . 24
15.5 Sweeps . . . 25
15.6 Tables in the dashboard . . . 25

16 Validation **26**

16.1 Against the datasheet and the chart points 26

16.2 Against the electrochemical model 27

17 Sensitivity to the uncertain inputs **27**

18 Automated tests **29**

19 Assumptions and limitations **29**

A Parameter table with sources **30**

B Where each formula lives in the code **32**

1 Model structure

The module has n_S series groups, each with n_P cells in parallel, so $N = n_S n_P$ cells. The default is $n_S = 4$, $n_P = 3$, $N = 12$. Cell $k = (g-1)n_P + i$ sits in series group $g = 1, \ldots, n_S$ at parallel position $i = 1, \ldots, n_P$. On the physical grid, series groups are columns and parallel positions are rows.

Five subsystems share one parameter set (`Design`, `Scenario`, `Requirements`):

1. **Electrical**: open-circuit voltage, a series resistance and one polarisation branch per cell, current sharing, charge and discharge limits (Section 4).
2. **Heat**: ohmic and polarisation losses per cell and in the interconnects (Section 5).
3. **Thermal**: a network of one node per cell plus an optional plate (Section 6).
4. **BMS**: voltage cut-offs and temperature trips (Section 7).
5. **Parts and checks**: mass, cost, volume, requirement checks and ranking (Sections 10–14).

A sixth layer, **liquid cooling** for Design D (Section 13), adds a cold plate, coolant, radiator and pump to the same thermal network. A seventh, **electrochemical** (Section 12), adds a lithium-plating screen taken from the PyBaMM Chen2020 model. It reads a stored table, so the platform runs without PyBaMM.

2 Notation

Symbol	Meaning	Unit or default
n_S, n_P, N	series groups, parallel cells per group, cell count	4, 3, 12
Q	cell capacity, defines 1C	4.85 Ah (DS 2.1, DS 2.9)
s	state of charge (SOC), 0 to 1	–
OCV(s)	open-circuit voltage of one cell	V, Table 3
R_0, R_1, τ	series resistance, polarisation resistance, polarisation time constant at 25 °C	24.78 mΩ, 14.07 mΩ, 65 s
C_1	polarisation capacitance, $C_1 = \tau / R_1$	4.62 kF

Symbol	Meaning	Unit or default
μ_k	resistance spread factor of cell k	–
$f(T)$	resistance multiplier vs temperature	–
$R_{0,k}, R_{1,k}$	$R_0\mu_k f(T_k)$ and $R_1\mu_k f(T_k)$	Ω
$V_{1,k}$	polarisation voltage across $R_1 \parallel C_1$ of cell k	V
R_{bus}	total interconnect resistance of the string	4 mΩ
I	module current, positive on discharge	A
I_k	current in cell k, same sign convention	A
$c_{\text{dis}}, c_{\text{chg}}$	discharge and charge rate in multiples of Q	C
T_k	temperature of node k	°C
T_{amb}	ambient temperature	35 °C
C_k	heat capacity of node k	J/K
$G_{\text{amb},k}$	conductance from node k to ambient	W/K
g_{nb}	conductance between neighbouring cells	W/K
h_{eff}	effective convection coefficient	W/m^2/K
Δt	time step	5 s (10 s in sweeps)

3 Cell data and their sources

All values are in Appendix A with their source tags. The ones that drive the formulas are: capacity $Q = 4.85$ Ah (minimum capacity; 1C = 4 850 mA, which is how the datasheet converts 0.3C to 1 455 mA, 0.7C to 3 395 mA and 1.5C to 7 275 mA), nominal voltage 3.63 V, charge voltage 4.20 ± 0.05 V, discharge cut-off 2.50 V, mass 68 g, diameter ≤ 21.10 mm, height ≤ 70.15 mm, DC resistance 30 ± 6 mΩ (0.5C, 30 s, 25 °C, 50 % SOC, DS 4.3.2), 1 kHz impedance ≤ 25 mΩ (DS 4.3.1).

	Charge	Discharge
Temperature window	0 to 45 °C (DS 5.1)	−20 to 60 °C (DS 5.1)
Maximum current	0.3C (1.455 A) from 0 to 25 °C; 0.7C (3.395 A) from 25 to 50 °C (DS 2.6)	0.3C (1.455 A) from −20 to 5 °C; 1.5C (7.275 A) from 5 to 60 °C (DS 2.9)
End condition	50 mA after a standard charge (DS 2.4); C/20 in the cycle-life test (DS 4.2.3)	2.50 V

DS 2.6 lists charge current up to 50 °C, while DS 5.1 limits charging to 0 to 45 °C. The model uses the stricter 45 °C. The datasheet gives no release temperature after a trip, so a 5 K hysteresis is assumed (charge releases at 40 °C, discharge at 55 °C).

4 Electrical sub-model

Source: `bms/cell.py`, `bms/sim.py`. Each cell is OCV(s) in series with $R_{0,k}$ and a parallel branch $R_{1,k} \parallel C_1$ (Figure 1). The branch stands for solid-state diffusion in the electrodes. It lengthens the

constant-voltage charge tail and lowers the energy delivered at high rate, which a resistor alone cannot do. It captures these effects only in part (Section 16).

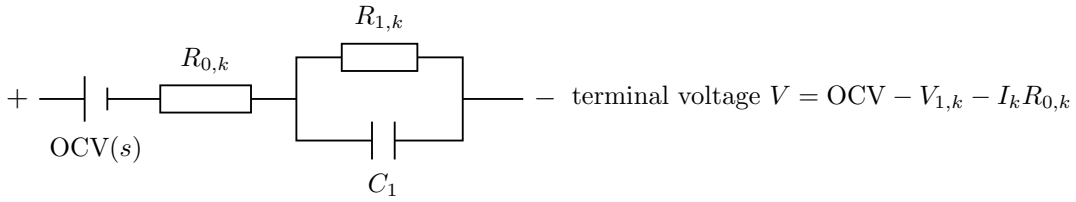


Figure 1: Equivalent circuit of one cell. $V_{1,k}$ is the voltage across the $R_1 \parallel C_1$ branch.

4.1 Open-circuit voltage

OCV(s) is piecewise linear through the 16 points in Table 3. They come from PB: a C/50 discharge of the `Chen2020` single-particle model at 25 °C, with the voltage read at equal steps of discharged capacity ($s = 1 -$ discharged fraction). The script `tools/make_ocv.py` regenerates them. This is a research parameterisation of an M50 cell, not a datasheet curve. Against the datasheet it gives $E = 17.8$ Wh at 0.2C (17.6 minimum, 18.2 nominal) and a mean discharge voltage of 3.68 V (3.63 nominal).

SOC (%)	0	2	5	10	15	20	30	40
OCV (V)	2.500	2.859	3.105	3.292	3.430	3.481	3.578	3.663
SOC (%)	50	60	70	80	90	95	98	100
OCV (V)	3.746	3.836	3.944	4.037	4.093	4.118	4.157	4.196

Table 3: Open-circuit voltage table (PB, 25 °C).

4.2 Resistance, temperature and cell spread

The two resistances of cell k are

$$R_{0,k} = R_0\,\mu_k\,f(T_k), \qquad R_{1,k} = R_1\,\mu_k\,f(T_k), \qquad \tau_k = R_{1,k}\,C_1, \tag{1}$$

so the capacitance C_1 is constant and the time constant scales with R_1. The multiplier $f(T)$ is read from the BD chart of DCIR (0.5 s pulse) against temperature, which has three clusters of points. The means are 68 mΩ at 10 °C (points 59, 70, 76), 23 mΩ at 25 °C (21, 23, 25) and 14.2 mΩ at 45 °C (14, 14.5). These were read from the chart by eye, with an uncertainty of about 2 mΩ. Dividing by the 25 °C mean gives $f(10) = 2.957$, $f(25) = 1$, $f(45) = 0.615$. Between the points $\ln f$ is interpolated linearly in T. Below 10 °C the 10–25 °C slope is extended, and above 45 °C the 45 °C value is held (so heat is not under-estimated when hot). Examples: $f(15) = 2.060$, $f(35) = 0.784$, $f(45) = 0.615$, $f(60) = 0.615$. Only the ratios are used; the absolute 25 °C value comes from DS 4.3.2.

The cell-to-cell spread is

$$\mu_k = \mathrm{clip}\left(1 + \sigma\,\xi_k,\ 0.7,\ 1.6\right), \qquad \xi_k \sim \mathcal{N}(0,1), \tag{2}$$

with $\sigma = 0.05$ by default and ξ_k drawn from a seeded generator (`numpy.random.default_rng(seed)`). The same seed gives the same result every run. The spread is applied to both R_0 and R_1.

4.3 Fitting R_0, R_1, τ

The datasheet gives only the 30 s resistance, so the split between R_0 and R_1 and the time constant come from a fit (`tools/fit_polarisation.py`):

1. In the PB DFN model at 25 °C, start at 50 % SOC, rest 10 min, discharge at 0.5C for 20 min, rest 30 min.
2. Fit the circuit voltage change to the DFN voltage change by least squares over the pulse and relaxation. Result: $R_0 = 29.89\,\mathrm{m\Omega}$, $R_1 = 16.98\,\mathrm{m\Omega}$, $\tau = 65\,\mathrm{s}$, rms error 3.0 mV. A second branch did not reduce the error, so one branch is used.
3. The DFN cell is more resistive than the datasheet: the fitted circuit gives a 30 s resistance of 36.2 mΩ (38.4 mΩ when read from the voltage as in DS 4.3.2, which includes the open-circuit-voltage drop over the 30 s) against 30 ± 6. Both R_0 and R_1 are multiplied by one factor, $30/36.2 = 0.829$, so that $R_0 + R_1(1 - e^{-30/\tau}) = 30\,\mathrm{m\Omega}$. This gives the values used: $R_0 = 24.78$, $R_1 = 14.07\,\mathrm{m\Omega}$.

The fit uses nothing from charge or high-rate tests, so those stay independent (Section 16). After scaling, the 0.5 s resistance is 24.9 mΩ, inside the 21 to 25 mΩ range of the BD points at 25 °C, and this was not fitted.

4.4 Polarisation state

With I_k positive on discharge, over a step Δt the branch is integrated exactly,

$$V_{1,k}^{n+1} = V_{1,k}^{n}\, e^{-\Delta t/\tau_k} + I_k\, R_{1,k}\left(1 - e^{-\Delta t/\tau_k}\right). \tag{3}$$

During charge $I_k < 0$, so $V_{1,k}$ is negative and adds to the terminal voltage.

4.5 Parallel groups and current sharing

Cells in group g share one terminal voltage V_g. From $I_{g,i} = (\mathrm{OCV} - V_{1,g,i} - V_g)/R_{0,g,i}$ and $\sum_i I_{g,i} = I$:

$$S_g = \sum_{i=1}^{n_\mathrm{P}} \frac{1}{R_{0,g,i}}, \quad U_g = \sum_{i=1}^{n_\mathrm{P}} \frac{V_{1,g,i}}{R_{0,g,i}}, \quad V_g = \underbrace{\mathrm{OCV} - \frac{U_g}{S_g}}_{A_g} - \underbrace{\frac{1}{S_g}}_{B_g} I. \tag{4}$$

A cell with lower R_0, including a hotter one, carries more current. All series groups carry the same module current. The module voltage is $V_\mathrm{mod} = \sum_g V_g - I R_\mathrm{bus}$.

4.6 Rates and state of charge

Rates are multiples of the cell capacity:

$$I_\mathrm{dis} = c_\mathrm{dis}\, n_\mathrm{P}\, Q, \qquad I_\mathrm{chg} = c_\mathrm{chg}\, n_\mathrm{P}\, Q. \tag{5}$$

At 1C with $n_\mathrm{P} = 3$: $I = 1 \times 3 \times 4.85 = 14.55\,\mathrm{A}$ in the module and 4.85 A in an average cell. The SOC is updated each step as

$$s_{n+1} = \mathrm{clip}\Big(s_n - \frac{I\,\Delta t}{3600\, n_\mathrm{P}\, Q},\ 0,\ 1\Big), \tag{6}$$

with I negative on charge. Coulombic efficiency is taken as 1. All cells share one SOC, so capacity mismatch is not modelled.

4.7 Discharge

Discharge ends when any of three conditions holds:

$$\min_g \frac{A_g - 2.50}{B_g} < I_{\text{dis}}, \qquad s \le 0, \qquad 1 - s \;\ge\; \rho(\bar{T}). \tag{7}$$

The first is the voltage cut-off at the requested current. The third uses the energy retention table of DS 4.4.4 (0.2C, charged at 25 °C): $\rho = 0.70, 0.80, 1.00, 0.95$ at $-10, 0, 25, 60$ °C, interpolated linearly in the mean cell temperature $\bar{T}$ and held outside the table. The resistance alone gives 94 % at 0 °C and 86 % at −10 °C (Section 16), so the table adds what the circuit cannot explain. At 35 °C it removes about 1.4 %.

4.8 Constant-current, constant-voltage charge

The largest charge current that keeps every group at or below 4.20 V is

$$I_V = \min_g \frac{4.20 - A_g}{B_g}. \tag{8}$$

The constant-current (CC) phase holds $I = I_{\text{chg}}$ until $I_V \le I_{\text{chg}}$. The constant-voltage (CV) phase then follows $I = \min(I_{\text{chg}}, \max(I_V, 0))$, so the current falls as the cells fill and V_1 relaxes. Charging stops when

$$I \le I_{\text{cut}} = n_{\text{P}}\, i_{\text{term}}, \qquad i_{\text{term}} = \begin{cases} 0.050 \text{ A} & i_{\text{chg,cell}} \le 1.455 \text{ A (DS 2.4)} \\ Q/20 = 0.2425 \text{ A} & \text{otherwise (DS 4.2.3)} \end{cases} \tag{9}$$

or when $s \ge 1$. For the default 0.6C charge $I_{\text{cut}} = 0.73$ A for the module. In rest phases $I = 0$ and the polarisation voltage relaxes with τ_k. Energy in a phase is

$$E = \sum_n V_{\text{mod},n}\, |I_n|\, \frac{\Delta t}{3600} \quad \text{[Wh]}. \tag{10}$$

4.9 Current limits as a check, not a clamp

The datasheet current limits depend on temperature (table in Section 3). The simulation does not clamp the current to them, so the checker can show a charger that does not match the cell. For each step the ratio

$$\rho_I = \max_k \frac{|I_k|}{I_{\text{lim}}\big(\,\text{clip}(T_k, T_{\min}, T_{\max})\big)} \tag{11}$$

is recorded, and checks L3 and L4 report its maximum. The temperature is clipped into the window so that ρ_I measures only the current-versus-band mismatch. Temperatures outside the window are reported by the temperature checks.

Why 0.6C is the default charge rate. The datasheet maximum is 0.7C, i.e. 3.395 A per cell. A charger set to 0.7C splits the current between parallel cells in proportion to $1/R_0$, and with a 5 % spread the most conductive cell carries about 4 % more than the 3.395 A, so L4 fails in every design. At 0.6C the same cell reaches 89 % of the limit.

5 Heat generation

Each cell produces

$$Q_k = I_k^2 R_{0,k} + \frac{1}{2}\,\frac{(V_{1,k}^n)^2 + (V_{1,k}^{n+1})^2}{R_{1,k}} + \frac{I^2 R_{\text{bus}}}{N}, \tag{12}$$

the loss in R_0, the loss in R_1 (average of the start and end of the step), and the interconnect loss spread equally over all cells. Entropic (reversible) heat is neglected because no M50 value is available in the sources (the PB set has it equal to zero). The module-level energy balance, terminal energy equal to OCV energy minus heat minus change of energy stored in C_1, is checked in `tests/test_model.py`.

Example at 1C in steady state, 35 °C (default 4S3P): $R_0 + R_1 = 30.46\,\mathrm{m\Omega}$ at $f = 0.784$, so $Q_{\mathrm{cell}} = 4.85^2 \times 0.03046 = 0.716\,\mathrm{W}$ per cell. The interconnect loss is $14.55^2 \times 0.004 = 0.847\,\mathrm{W}$ for the string. The module total is $12 \times 0.716 + 0.847 = 9.44\,\mathrm{W}$. During the first minutes V_1 has not built up and the loss is lower, which is why the temperature curve starts with a gentler slope.

6 Thermal network

Source: ***`build_thermal`*** *and* ***`simulate`*** *in* ***`bms/sim.py`****.* There is one temperature node per cell, plus one node for the aluminium plate if the design has one.

6.1 Heat capacity

$$C_k = m_{\mathrm{cell}}\, c_p = 0.068 \times 900 = 61.2\ \mathrm{J/K}, \qquad C_{\mathrm{sp}} = \rho_{\mathrm{Al}}\, A_{\mathrm{sp}}\, t_{\mathrm{sp}}\, c_{\mathrm{Al}}. \tag{13}$$

The cell specific heat $c_p = 900$ J/kg/K is an assumption: typical cylindrical Li-ion cells fall between 800 and 1100, and neither the specification nor the BD page gives an M50 value. The PB layer values (700 J/kg/K for electrodes and separator, 385 and 897 for the collectors) cover the jelly roll only, not the steel can. The sensitivity run varies it. For the plate, $\rho_{\mathrm{Al}} = 2700\ \mathrm{kg/m^3}$, $c_{\mathrm{Al}} = 900$ J/kg/K. With $A_{\mathrm{sp}} = 125\ \mathrm{cm^2}$ and $t_{\mathrm{sp}} = 2$ mm (design B) the plate weighs 67.5 g and $C_{\mathrm{sp}} = 60.8$ J/K.

6.2 Convection to ambient

The effective coefficient is

$$h_{\mathrm{eff}} = h_{\mathrm{nat}} + \begin{cases} k_f\, v^{0.6} & v > 0 \\ 0 & v = 0 \end{cases}, \qquad h_{\mathrm{nat}} = 10,\ k_f = 12. \tag{14}$$

Natural convection plus radiation $h_{\mathrm{nat}} = 10\ \mathrm{W/m^2/K}$ is an assumption. The forced-air term is a bank-derated power law (assumption): a 2 m/s fan gives $h_{\mathrm{eff}} = 10 + 12 \times 2^{0.6} = 28.2\ \mathrm{W/m^2/K}$.

Each cell has four vertical sides (north, south, east, west on the grid). The surface area of one side is

$$a_{\mathrm{side}} = \frac{\pi D L}{4} = \frac{\pi \times 0.0211 \times 0.07015}{4} = 11.63\ \mathrm{cm^2}. \tag{15}$$

A side facing a neighbour is only partly open to the air. Its exposure depends on the gap s:

$$\varphi(s) = 1 - \exp\!\left(-\frac{s}{s_0}\right), \qquad s_0 = 3\ \mathrm{mm}\ \text{(assumption)}. \tag{16}$$

So $\varphi(1.5\ \mathrm{mm}) = 0.393$ and $\varphi(4\ \mathrm{mm}) = 0.736$. With $n_{b,k}$ the number of neighbouring cells of cell k (4-connected grid), the conductance to ambient is

$$G_{\mathrm{amb},k} = h_{\mathrm{eff}}\, a_{\mathrm{side}} \Big[(4 - n_{b,k}) + \varphi(s)\, n_{b,k} \Big]. \tag{17}$$

Corner cells have 2 neighbours, edge cells 3, and the two interior cells of a 4×3 grid have 4. Cell tops and bottoms are treated as insulated.

G_{amb} (W/K)	Corner	Edge	Interior
A, B (1.5 mm gap, natural convection)	0.0324	0.0253	0.0183
C (4 mm gap, 2 m/s fan)	0.1138	0.1052	0.0965

Table 4: Conductance from each cell to ambient. Interior cells cool worst, which creates the hotspot.

6.3 Conduction between neighbouring cells

Through the air gap, with an effective contact width of $0.4D$ (assumption) and a minimum gap of 0.5 mm:

$$g_{\text{nb}} = \frac{k_{\text{air}}\, L\,(0.4D)}{\max(s,\, 0.5\ \text{mm})}, \qquad k_{\text{air}} = 0.026\ \text{W/m/K}. \tag{18}$$

This gives 0.0103 W/K at 1.5 mm and 0.0038 W/K at 4 mm.

6.4 Spreader plate (design B)

Each cell connects to the plate through a thermal pad, and the plate sheds heat from both faces:

$$g_{cs} = \frac{k_{\text{pad}}\, A_{\text{pad}}}{t_{\text{pad}}} = \frac{1.5 \times 5 \times 10^{-4}}{1 \times 10^{-3}} = 0.75\ \text{W/K}, \qquad G_{\text{sp,amb}} = 2\, h_{\text{eff}}\, A_{\text{sp}} = 2{\times}10{\times}0.0125 = 0.25\ \text{W/K}. \tag{19}$$

The pad conductivity 1.5 W/m/K and the pad geometry are assumptions. The plate area (125 cm^2, about twice the 61 cm^2 footprint of the cell block) and the pad contact patch (5 cm^2 per cell) are design choices fixed before the sensitivity run; Section 17 varies the pad conductivity.

6.5 Governing equation and solver

Collecting all nodes into vectors,

$$\mathbf{C}\,\frac{d\mathbf{T}}{dt} = -\mathbf{K}\,\mathbf{T} + \mathbf{Q} + \mathbf{G}_{\text{amb}}\,T_{\text{amb}}, \qquad \mathbf{K} = \mathbf{L} + \text{diag}(\mathbf{G}_{\text{amb}}), \tag{20}$$

where $\mathbf{L}$ is the weighted graph Laplacian built from g_{nb} and g_{cs} ($L_{kk} = \sum_j g_{kj}$, $L_{kj} = -g_{kj}$). The plate node has no heat source. Backward Euler is used,

$$\left(\tfrac{\mathbf{C}}{\Delta t} + \mathbf{K}\right)\mathbf{T}^{n+1} = \tfrac{\mathbf{C}}{\Delta t}\mathbf{T}^{n} + \mathbf{Q}^{n} + \mathbf{G}_{\text{amb}}T_{\text{amb}}. \tag{21}$$

The matrix depends only on the design, so it is inverted once per run. The scheme is stable for any step size, which matters because the plate node has a time constant of a few seconds.

6.6 Time constants and a hand check

For a lumped estimate, $\tau_{\text{th}} = \sum C / \sum G_{\text{amb}}$ and the long-time rise is $\Delta T_\infty = Q_{\text{tot}} / \sum G_{\text{amb}}$.

Design	$\sum G_{\text{amb}}$ (W/K)	τ_{th}	ΔT_∞ at 35 °C	Simulated mean cell, end of discharge
A	0.3183	2307 s (38 min)	29.7 K	53.8 °C
B	0.5683	1399 s (23 min)	16.6 K	47.5 °C
C	1.2793	162 s (10 min)	7.4 K	41.4 °C

For design A at 1C: $Q_{\mathrm{tot}} = 9.44\,\mathrm{W}$ gives $\Delta T_\infty = 29.7\,\mathrm{K}$ when R is taken at 35 °C. At a cell temperature near 50 °C the resistance is lower ($f = 0.615$), $Q_{\mathrm{tot}} = 7.59\,\mathrm{W}$ and $\Delta T_\infty = 23.8\,\mathrm{K}$. A first-order rise after $t = 58\,\mathrm{min}$ covers $1 - e^{-t/\tau_{\mathrm{th}}} = 0.78$ of the way, so the mean cell temperature should be about 54 to 58 °C. The simulated mean at the end of the discharge is 53.8 °C. This is a sanity check on the thermal solver, not a validation against a measurement.

7 BMS and protection logic

The BMS sees the temperature of one sensor cell, T_s: cell 0 (a corner) for `corner`, cell $\lfloor n_{\mathrm{S}}/2 \rfloor n_{\mathrm{P}} + \lfloor n_{\mathrm{P}}/2 \rfloor$ for `center`, or $\max_k T_k$ for `hottest` (ideal sensing).

Mode	Trip when	Release when	Source
Discharge	$T_s \geq 60\,^\circ\mathrm{C}$	$T_s \leq 55\,^\circ\mathrm{C}$	DS 5.1, release assumed
Charge	$T_s \geq 45\,^\circ\mathrm{C}$	$T_s \leq 40\,^\circ\mathrm{C}$	DS 5.1, release assumed
Charge inhibit	$\min_k T_k < 0\,^\circ\mathrm{C}$	automatic	DS 5.1
Discharge end	$\min_g (A_g - 2.5)/B_g < I_{\mathrm{dis}}$	–	DS 2.7
Charge CC to CV	$I_V \leq I_{\mathrm{chg}}$	–	DS 2.5

When tripped the current is set to zero and the time is counted as trip time. The charge cut-off acts on the sensor reading, not on the hottest cell. With the corner sensor under-reading the hotspot by several kelvin, the BMS can allow charging while the hottest cell is above 45 °C. Design A shows this: the sensor holds the string near 45 °C, and the hottest cell reaches 53.1 °C during charge (check L1). The BMS also trips only at datasheet limits, so a design can break the application requirement without the BMS reacting (check I2).

8 Duty cycle and time stepping

The run starts fully charged at ambient temperature and repeats *discharge, rest, charge, rest* for `n_cycles` cycles. Rest lasts 10 min. Each phase is capped at 8 h. At every step Δt:

1. Compute $R_{0,k}$, $R_{1,k}$, A_g, B_g from the current temperatures and polarisation states.
2. Read the sensor temperature and update the trip state.
3. Test the phase end conditions (Sections 4 and 7). Set the module current I.
4. Compute cell currents I_k, heat Q_k, group voltages and V_{mod}. Record the step, including the limit ratio ρ_I.
5. Add $V_{\mathrm{mod}}|I|\Delta t/3600$ to the phase energy, update s and $V_{1,k}$.
6. Solve the thermal step for $\mathbf{T}^{n+1}$ and advance time.

Results are always taken from the first discharge and first charge (cold start) for energy and charge time, and over the whole run for peak temperatures. The second cycle starts warm, which is why the peak occurs late.

9 Indicators

Source: `indicators` *in* `bms/sim.py`. Let $T_{\max}(t) = \max_k T_k(t)$ and $T_{\min}(t) = \min_k T_k(t)$ over the cell nodes only (the plate node is excluded).

$E_{\text{use}} = E_{\text{dis},1} - P_{\text{fan}} \dfrac{t_{\text{dis},1}}{3600}$	usable energy, net of fan energy, first discharge
$T_{\text{peak}} = \max_t T_{\max}(t)$	peak cell temperature, whole run
$\Delta T_{\text{spread}} = \max_t \left[T_{\max}(t) - T_{\min}(t)\right]$	largest temperature difference between cells
$t_{\text{chg}} =$ duration of the first charge phase	includes the CV tail and BMS pauses
$\eta_{\text{rt}} = E_{\text{dis},1}/E_{\text{chg},1}$	round trip, SOC returns to 100 %, fan excluded
$I_{\text{cell,max}} = \max_t \max_k \lvert I_k(t)\rvert$	reported separately for discharge and charge
$\rho_I = \max_t \max_k \lvert I_k\rvert / I_{\text{lim}}(T_k)$	limit ratio, discharge and charge
$V_{g,\min} = \min_{t,g} V_g(t)$	lowest group voltage
$\delta_{\text{sensor}} = \max_t \left[T_{\max}(t) - T_s(t)\right]$	how much the sensor under-reads the hotspot
$\rho_E = E_{\text{use}}/m, \qquad \rho_V = E_{\text{use}}/(V/1000)$	Wh/kg and Wh/L

Module voltages: $V_{\text{nom}} = 3.63\,n_{\text{S}}$, $V_{\min} = 2.5\,n_{\text{S}}$, $V_{\max} = 4.2\,n_{\text{S}}$. For the default module the delivered energy at 1C is 202.3 Wh. The datasheet nominal value $N\,Q\,3.63 = 211.3$ Wh is for a 0.2C discharge at 25 °C; at 1C and 35 °C the polarisation drop and the retention factor lower it.

Indicator	A Baseline	B Spreader	C Fan, wide gaps
Usable energy (Wh)	202.3	203.6	203.9
Peak cell temperature (°C)	57.3	47.7	41.8
Temperature spread (K)	4.67	0.21	0.82
Charge time (min)	136.5	103.7	105.2
Discharge time (min)	57.6	58.1	58.7
Round-trip efficiency	0.946	0.945	0.942
Mass (g)	878	952	890
Cost (INR)	6226	6586	6376
Volume (cm^3)	430	467	550
Energy density (Wh/kg)	230.4	214.0	229.1
Energy density (Wh/L)	470	436	371
Sensor under-read (K)	4.67	0.21	0.77
Peak charge current vs limit (%)	89	89	89

Table 5: Indicators at the default scenario (35 °C, 1C discharge, 0.6C charge, two cycles).

10 Mass, cost and volume

Source: `bms/design.py`. *All non-cell values are assumptions.*

$$m\ [\mathrm{g}] = N\,(68+2) + \big(10 + 4\,n_\mathrm{S}\big) + 12 + \mathbb{1}_\mathrm{sp}\big(1000\,\rho_\mathrm{Al} A_\mathrm{sp} t_\mathrm{sp} + 0.5\,N\big) + 12\,\mathbb{1}_\mathrm{fan} \tag{22}$$

$$c\ [\mathrm{INR}] = N\,(450+8) + \big(150 + 20\,n_\mathrm{S}\big) + \big(200 + 75\,n_\mathrm{S}\big) + \mathbb{1}_\mathrm{sp}\big(0.3\,m_\mathrm{sp} + 100 + 20\,N\big) + 150\,\mathbb{1}_\mathrm{fan} \tag{23}$$

$$V\ [\mathrm{cm}^3] = \frac{(n_\mathrm{S}\,p)(n_\mathrm{P}\,p)\,L}{1000} + \mathbb{1}_\mathrm{sp}\,0.3\,A_\mathrm{sp} + 20\,\mathbb{1}_\mathrm{fan}, \qquad p = D + s\ [\mathrm{mm}] \tag{24}$$

Terms, in the order written: cells (68 g, DS 2.10) with 2 g holder each, nickel strip and wiring, BMS board, plate with 0.5 g pad per cell, 40 mm fan. For cost: cell price (INR 450, placeholder) with an INR 8 holder, strip and wiring, BMS board that grows with series count, plate material at INR 0.3 per gram plus machining and pads, and the fan. For volume, L is the cell height in mm, the plate adds a 3 mm stack over its area, and the fan with duct adds 20 cm^3. $\mathbb{1}$ is 1 if the part is present and 0 otherwise.

	A	B	C
Cells and holders (g)	840	840	840
Strip, wiring, BMS (g)	38	38	38
Plate and pads (g)	0	73.5	0
Fan (g)	0	0	12
Mass (g)	**878**	**951.5**	**890**
Cells and holders (INR)	5496	5496	5496
Strip and wiring (INR)	230	230	230
BMS board (INR)	500	500	500
Plate, machining, pads (INR)	0	360.2	0
Fan (INR)	0	0	150
Cost (INR)	**6226**	**6586.2**	**6376**
Volume (cm^3)	**430.0**	**467.5**	**550.3**

Example for A: pitch $p = 21.1 + 1.5 = 22.6$ mm, footprint $4p \times 3p = 90.4 \times 67.8$ mm, volume $90.4 \times 67.8 \times 70.15/1000 = 430.0\,\mathrm{cm}^3$.

11 Checks

Source: `run_checks` *in* `bms/checks.py`. Each check compares a simulated value with a limit and returns PASS or FAIL.

ID	Quantity	Pass condition	Basis
L1	peak temperature during charge	$\leq 45\,^\circ\mathrm{C}$	DS 5.1
L2	peak temperature during discharge	$\leq 60\,^\circ\mathrm{C}$	DS 5.1
L3	ρ_I during discharge	$\leq 100\,\%$	DS 2.9
L4	ρ_I during charge	$\leq 100\,\%$	DS 2.6

ID	Quantity	Pass condition	Basis
L5	lowest group voltage	$\geq 2.5\,\mathrm{V}$	DS 2.7
L6	ambient temperature	0 to 45 °C	DS 5.1 no charging outside
R1	T_{peak}	$\leq T_{\mathrm{req}}$ (45 °C)	top of the datasheet charge window: above it the BMS blocks charging
R2	E_{use}	$\geq 190\,\mathrm{Wh}$	requirement (editable)
R3	t_{chg}	$\leq 120\,\mathrm{min}$	requirement
R4	mass	$\leq 1100\,\mathrm{g}$	requirement
R5	cost	$\leq$ INR 8000	requirement
R6	volume	$\leq 600\,\mathrm{cm}^3$	requirement
R7	$[V_{\mathrm{min}}, V_{\mathrm{max}}]$	inside the load window (10 to 16.8 V)	series count sets the window
R8	gap between cells	$\geq 1\,\mathrm{mm}$	design rule; the datasheet gives no spacing
E1	ρ_{pl} (Section 12)	$\leq 100\,\%$	plating screen, Chen2020 DFN at 25 °C
LQ1–LQ4	liquid loop (Section 13)	see Section 13	design D only
I1	$c_{\mathrm{chg}}Q$	$\leq I_{\mathrm{lim}}(T_{\mathrm{amb}})$	a cold cell accepts only 0.3C
I2	T_{peak} vs BMS trip level	$T_{\mathrm{peak}} \leq T_{\mathrm{req}}$	discharge trip at 60 is 15 K above the requirement, so a miss raises no alarm
I3	δ_{sensor}	$\leq 2\,\mathrm{K}$	sensor must see the hotspot
I4	$100\,P_{\mathrm{fan}}/\left(E_{\mathrm{use}}/t_{\mathrm{dis}}\right)$	$\leq 2\,\%$	parasitic load on module power

The requirement values are editable defaults chosen for a 14.5 V, 200 Wh class module, not figures from the problem statement. The traceability table (Section 15) links each requirement to the component, assumption, model, result and evidence. It is built from the same results, so it updates with every change.

12 Electrochemical layer: lithium-plating screen

Source: `bms/echem.py`, `tools/make_plating_map.py`, `tools/echem_replay.py`, `data/plating_map.csv`.

Why it is there. The equivalent circuit knows terminal voltage, heat and temperature. It has no electrode variables, so it cannot say whether a charge current is safe for the anode. The PB electrochemical model can. Its negative-electrode potential against lithium metal, at the separator side, is ϕ_{n}. If $\phi_{\mathrm{n}} < 0\,\mathrm{V}$, lithium plating is thermodynamically possible. This is a screening criterion. It does not give the amount of plated lithium.

Plating map. `tools/make_plating_map.py` charges the DFN (and the faster SPMe for comparison) from empty at 25 °C at constant current $c = 0.1, \ldots, 1.5$ (datasheet C, 1 C = 4.85 A) until 4.2 V, and stores ϕ_n against state of charge. Where a rate reaches 4.2 V before a given SOC, its last value is held. Table 7 gives the lowest potential reached at each rate.

Charge rate (C)	0.3	0.5	0.6	0.7	0.8	1	1.5
DFN, lowest ϕ_n (mV)	39	16	8	0	-6	-15	-47
SPMe, lowest ϕ_n (mV)	40	16	8	-2	-11	-25	-40

Table 7: Lowest negative-electrode potential during CC charge from empty, Chen2020, 25 °C.

Plating-free current. For each SOC x the limit $c_{pl}(x)$ is the highest mapped rate with $\phi_n \geq 0$, interpolated linearly in rate between the two mapped rates that bracket the zero crossing, and capped at 1.5C:

$$I_{pl}(x) = c_{pl}(x)\, Q, \qquad \rho_{pl} = \max_{t \in \text{charge}} \; \max_{k:\, T_k \geq 25\,^\circ\text{C}} \frac{|I_k(t)|}{I_{pl}\big(x(t)\big)}. \tag{25}$$

I_k is the current of cell k from the circuit, so the spread between parallel cells enters. The result is the indicator `peak_plating_ratio` and check E1 ($\rho_{pl} \leq 100\,\%$). At 25 °C the DFN limit is 1.17 C at 50 % SOC and 0.71 C at 90 % SOC.

What the map says about the datasheet. Above about 80 % SOC the DFN zero crossing sits at about 0.7C, which is where DS 2.6 puts the 25 to 45 °C charge limit. This is a consistency check on the datasheet limit, not a proof of it. Below 80 % SOC the DFN allows more than 0.7C, but the datasheet limit still applies, so the electrochemical limit is never used to raise the charge current.

Replay. `tools/echem_replay.py` runs the circuit at 25 °C ambient (design C, one cycle), takes the charge current of the most heavily loaded cell, and applies it to the DFN through a time-interpolated current function. Table 8 compares the two models and Figure 2 shows the curves.

Charge rate	Peak cell current (A)	Offset circuit − DFN, CC (mV)	DFN lowest ϕ_n (mV)	Time below 0 mV (min)	Platform ratio ρ_{pl} (%)
0.6C	3.03	-61	6.4	0.0	88
0.8C	4.04	-81	-7.7	2.8	105

Table 8: Platform cell current replayed through the Chen2020 DFN, 25 °C.

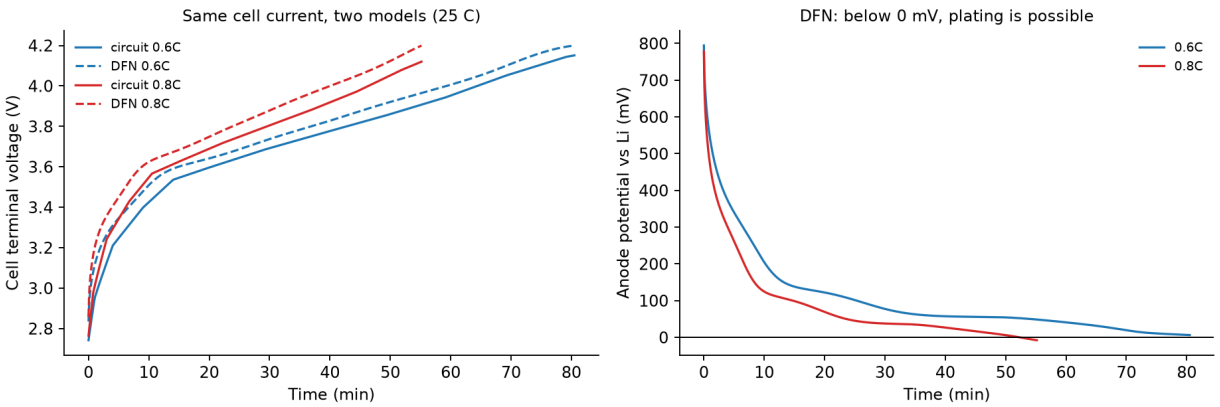


Figure 2: Left: terminal voltage of the circuit and of the DFN for the same cell current. Right: DFN negative-electrode potential.

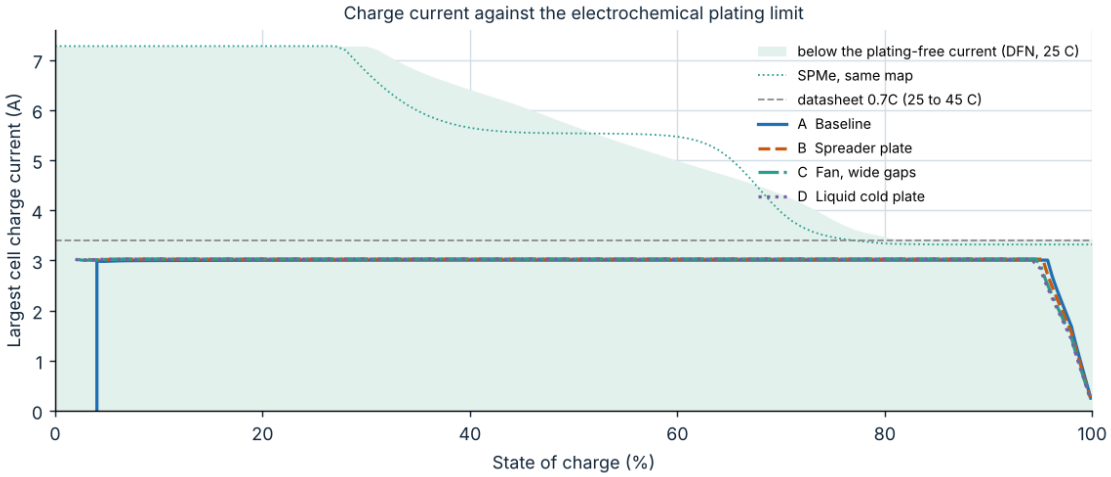


Figure 3: Largest cell charge current of the three designs during the first charge (35 °C ambient, 0.6C) against the plating-free current.

Reading. At 0.6C the DFN stays above 0 mV and the platform ratio is below 100 %. At 0.8C the DFN goes below 0 mV and the platform ratio is above 100 %, so the map-based check and the direct DFN replay agree on both cases. The circuit voltage is below the DFN voltage during the CC phase because the Chen2020 cell is about 28 % more resistive than the datasheet cell (38 against 30 mΩ at 30 s).

Limits of this layer.

- The Chen2020 set has no temperature dependence (Section 16). The screen is evaluated only for cells at 25 °C or above, and the charge time spent below 25 °C is reported (`plating_unrated_s`). Plating risk rises when the cell is colder, so the screen says nothing about cold charging. The datasheet 0.3C limit below 25 °C stays the controlling rule there.
- The map is for CC charge from empty. Charging from a partly charged state, or after a rest, changes the concentration gradients.
- SOC is rated-capacity based in the circuit and accepted-charge based in the map (4.89 Ah on the map scale). They differ by about 1 %.
- The electrochemical layer does not feed temperature, energy or charge time. Those still come from the circuit and thermal network.
- The set is a research fit of one M50 cell, not an LG model.

13 Liquid cooling: Design D

Source: `liquid_parts`, `build_thermal` *and* `simulate` *in* `bms/sim.py`; `Design` *and the parts roll-up in* `bms/design.py`; *checks LQ1 to LQ4 in* `bms/checks.py`; `bms/study.py`.

Purpose and status. Design D asks whether an aluminium cold plate with a water-glycol loop and a small radiator controls cell temperature better than designs A, B and C. It is added to the same thermal network. Designs A, B and C are untouched: their results are identical to the values in the earlier sections (tested in `tests/test_model.py`). Every liquid-loop parameter is an engineering estimate or a placeholder. **None of it is a measurement or a manufacturer specification, and the model has not been validated against a test.**

13.1 Nodes and heat paths

Two nodes are added after the N cell nodes: the cold plate p (temperature T_p, heat capacity $C_p = m_p c_{p,\mathrm{Al}}$) and the coolant c (temperature T_c, heat capacity $C_c = \rho V c_{p,\mathrm{cool}}$). The paths are

1. cell k to plate through its thermal pad, $G_{\mathrm{pad}} = k_{\mathrm{pad}} A_{\mathrm{pad}} / t_{\mathrm{pad}}$ (the same expression as the spreader design),
2. plate to coolant, with a heat-exchanger conductance UA_p,
3. coolant transport around the loop, $\dot{Q} = \dot{m}\, c_{p,\mathrm{cool}}\, (T_{\mathrm{out}} - T_{\mathrm{in}})$,
4. coolant to ambient air through the radiator, with conductance UA_r,
5. the cells still lose a little heat to the air from their sides, as in design A.

The coolant capacity rate is $\dot{C} = \dot{m}\, c_{p,\mathrm{cool}}$ (W/K). It sets how much of each conductance is actually available at a given flow.

13.2 Heat-exchanger conductances

For a fluid stream of capacity rate $\dot{C}$ exchanging heat with a body held at a fixed temperature through a conductance UA, the heat transferred per kelvin of temperature difference is

$$\varepsilon_x = 1 - \exp\Big(-\frac{UA_x}{\dot{C}}\Big), \qquad G_x = \dot{C}\,\varepsilon_x, \qquad x \in \{p,\ r\}. \tag{26}$$

This is the effectiveness–NTU result for the case where the other side has a much larger capacity rate (the plate and the air stream). It is an initial lumped approximation, not a CFD result. With no flow, $\dot{C} = 0$ and both conductances are zero, so a stopped pump isolates the plate from the coolant. The radiator conductance with its fan stopped is UA_r multiplied by 0.25 (ASSUMPTION).

13.3 Governing equations

The radiator returns the coolant to the plate inlet at

$$T_{\mathrm{in}} = T_c - \varepsilon_r\,(T_c - T_{\mathrm{amb}}), \qquad Q_p = G_p\,(T_p - T_{\mathrm{in}}), \qquad T_{\mathrm{out}} = T_{\mathrm{in}} + \frac{Q_p}{\dot{C}}, \tag{27}$$

so $Q_p = \dot{m}\, c_{p,\mathrm{cool}}(T_{\mathrm{out}} - T_{\mathrm{in}})$ holds by construction. The two new node equations are

$$C_p \frac{dT_p}{dt} = \sum_k G_{\mathrm{pad}}(T_k - T_p) - Q_p - G_{\mathrm{air}}(T_p - T_{\mathrm{amb}}), \tag{28}$$

$$C_c \frac{dT_c}{dt} = Q_p - G_r\,(T_c - T_{\mathrm{amb}}) + \eta_{\mathrm{pump}} P_{\mathrm{pump}}. \tag{29}$$

The cell equations are those of Section 6 with one extra term, $-G_{\mathrm{pad}}(T_k - T_p)$. The pump heat enters the coolant at the radiator inlet with $\eta_{\mathrm{pump}} = 1$ (ASSUMPTION: all electrical power ends as heat in the fluid). The radiator fan power is not added as heat because it heats the air stream. In matrix form the network is $C\,\dot{T} = -KT + w\,T_{\mathrm{amb}} + q$ as before, with K not symmetric because heat is carried by the flow. The same backward Euler step and the same precomputed inverse are used. A second inverse, with the pump stopped, is used during rest phases (when the controller stops the pump and radiator fan) and after a pump fault.

Steady state. Setting the derivatives to zero and eliminating T_{in} gives a series conductance between the plate and ambient for heat from the cells,

$$G_{\mathrm{eff}} = \dot{C}\,\frac{\varepsilon_r \varepsilon_p}{\varepsilon_r + \varepsilon_p - \varepsilon_r \varepsilon_p} \ \longrightarrow\ \frac{UA_r UA_p}{UA_r + UA_p} \quad (\dot{C} \to \infty). \tag{30}$$

With the default values $\dot{C} = 100.5\,\mathrm{W/K}$, $\varepsilon_r = 0.077$, $\varepsilon_p = 0.058$, so $G_{\mathrm{eff}} = 3.43\,\mathrm{W/K}$ against $3.43\,\mathrm{W/K}$ at infinite flow. Cells cannot be cooled below ambient: $T_c \geq T_{\mathrm{amb}}$ in every case, which the tests check. The plate-to-ambient temperature rise for about 9 W of heat is therefore roughly 2.6 K, and the pad and spread add the rest.

13.4 Auxiliary energy, heat and net usable energy

Pump and radiator fan power are electrical loads. The auxiliary energy of a run is $E_{\mathrm{aux}} = \int (P_{\mathrm{pump}} + P_{\mathrm{fan,rad}})\,dt$, and the net usable energy of the first discharge is

$$E_{\mathrm{use}} = E_{\mathrm{dis}} - \int_{\mathrm{discharge}} (P_{\mathrm{pump}} + P_{\mathrm{fan,rad}})\,dt, \tag{31}$$

the same rule as the fan of design C. Check I4 compares pump and fan power with module power. The pump heat enters the thermal network and its electrical energy enters E_{aux}; these are two views of one energy and are not counted twice in either place.

Energy conservation. Heat made in the cells and interconnects plus pump heat equals the heat stored in cells, plate and coolant plus the heat sent to ambient (air convection and radiator). For the default two-cycle run this closes to 0.006 Wh out of 49.3 Wh, and the coolant node balances at every step to 10^{-11} W away from the steps where the pump switches (`tests/test_model.py`).

13.5 Checks added for the liquid loop

ID	Quantity	Pass condition	Basis
LQ1	coolant rise across the plate, $T_{\mathrm{out}} - T_{\mathrm{in}}$	$\leq 5\,\mathrm{K}$	flow adequacy, design guideline
LQ2	$G_r(T_{\mathrm{req}} - T_{\mathrm{amb}})$ against the mean heat load (cells, bus, pump)	$\geq 100\,\%$	the radiator can hold the coolant at the requirement; zero at $T_{\mathrm{amb}} \geq T_{\mathrm{req}}$
LQ3	hottest cell minus coolant	$\leq 10\,\mathrm{K}$	share used by pad, plate and plate-to-coolant resistance
LQ4	pad is an electrical insulator	yes	design rule, cannot be simulated; cell cans sit at different potentials

The 5 K and 10 K limits are guidelines, not datasheet values. The ranking counts liquid-loop fails after requirement fails.

13.6 Parameters and sources

Parameter	Value	Unit	Source
Coolant type	Water-glycol 50/50	-	preset names a typical mixture
Coolant specific heat	3350	J/kg/K	ESTIMATE: typical handbook value near 30 to 40 C, not a supplier specification
Coolant density	1065	kg/m3	ESTIMATE: typical handbook value, not a supplier specification
Coolant volume in the whole loop	0.35	L	ESTIMATE: plate channels, hoses, radiator, reservoir
Coolant mass flow rate	0.03	kg/s	ESTIMATE: small DC pump class, oversized for about 9 W of heat
Coolant temperature at the start	ambient	C	input: clipped to ambient or above (no refrigeration)
Cold plate area	80	cm2	ESTIMATE: 4S3P footprint is about 61 cm2
Cold plate thickness	6	mm	ESTIMATE: milled channels inside
Cold plate mass	0.085	kg	ESTIMATE: area x thickness x 2700 kg/m3 x 65 % solid fraction
Cold plate to coolant conductance UA	6	W/K	ESTIMATE: laminar channel, h about 400 W/m2/K x 0.015 m2 wetted
Radiator UA, fan on	8	W/K	ESTIMATE: compact radiator with an 80 to 92 mm fan, no supplier data
Pump electrical power	4	W	ESTIMATE: small brushless DC pump class
Radiator fan electrical power	2	W	ESTIMATE: 80 to 92 mm fan class
Thermal pad conductivity	1.5	W/m/K	ASSUMPTION: shared with the spreader design
Thermal pad thickness	1	mm	ASSUMPTION
Pad contact area per cell	3.5	cm2	cell base area, 21.1 mm diameter (DS 3.2)
Pump, radiator, fan, hoses, fittings: mass	250	g	PLACEHOLDER: weigh the real parts
Same parts: packaged volume	250	cm3	PLACEHOLDER
Loop hardware cost	1650	INR	PLACEHOLDER, not a quote. Replace with supplier prices
Radiator UA without its fan, share of the fan-on value	0.25	-	ASSUMPTION: sim.RAD_NATURAL_FRACTION
Pump heat entering the coolant, share of pump power	1.0	-	ASSUMPTION: wet-rotor pump, sim.PUMP_HEAT_FRACTION

Table 10: Design D parameters. ESTIMATE and PLACEHOLDER values are not measurements or supplier specifications.

13.7 Result: four cooling systems, default case

35 °C ambient, 1C discharge, 0.6C charge, two cycles.

	A	B	C	D
Peak cell temperature (°C)	57.3	47.7	41.8	38.8
Average cell temperature (°C)	44.8	42.1	38.5	37.3
Cell-to-cell spread (K)	4.67	0.21	0.82	0.12
Net usable energy (Wh)	202.3	203.6	203.9	198.9
Charge time (min)	136	104	105	106
Round-trip efficiency (%)	94.6	94.5	94.2	94.0
Auxiliary energy, whole run (Wh)	0.0	0.0	4.9	32.9
pump (Wh)	0.0	0.0	0.0	21.9
radiator fan (Wh)	0.0	0.0	0.0	11.0
module fan (Wh)	0.0	0.0	4.9	0.0
Mass (g)	878	952	890	1592
Estimated cost (INR, placeholders)	6226	6586	6376	8116
Volume (cm^3)	430	467	550	728
Energy density (Wh/kg)	230	214	229	125
Energy density (Wh/L)	470	436	371	273
Fails: datasheet, requirement, interface, liquid-loop	1, 2, 2, 0	0, 1, 1, 0	0, 0, 0, 0	0, 3, 1, 0
Weighted score (four designs)	0.500	0.789	0.890	0.440

Table 11: Comparison of the four cooling systems. Score is the min–max weighted score over the four designs.

Reading. Design D has the lowest and most uniform temperatures (38.8 °C peak against 41.8 for C, spread 0.12 K), but it is the heaviest, largest and most expensive design (1592 g, 728 cm^3, INR 8116, all from placeholder part values) and it spends 32.9 Wh on pump and fans over the two-cycle run, which lowers net usable energy by 5.9 Wh in the first discharge. It fails the mass, cost and volume requirements and the fan-and-pump share check, so it ranks 3 of 4 and the ranking recommends C Fan, wide gaps. Design C already meets the 45 °C requirement at this operating point, so the extra 3 K of margin does not pay for itself here. With the requirement limits of mass, volume and cost relaxed, D would rank second behind C in this case; the ranking is a result of the simulation and of the editable limits, not a built-in preference.

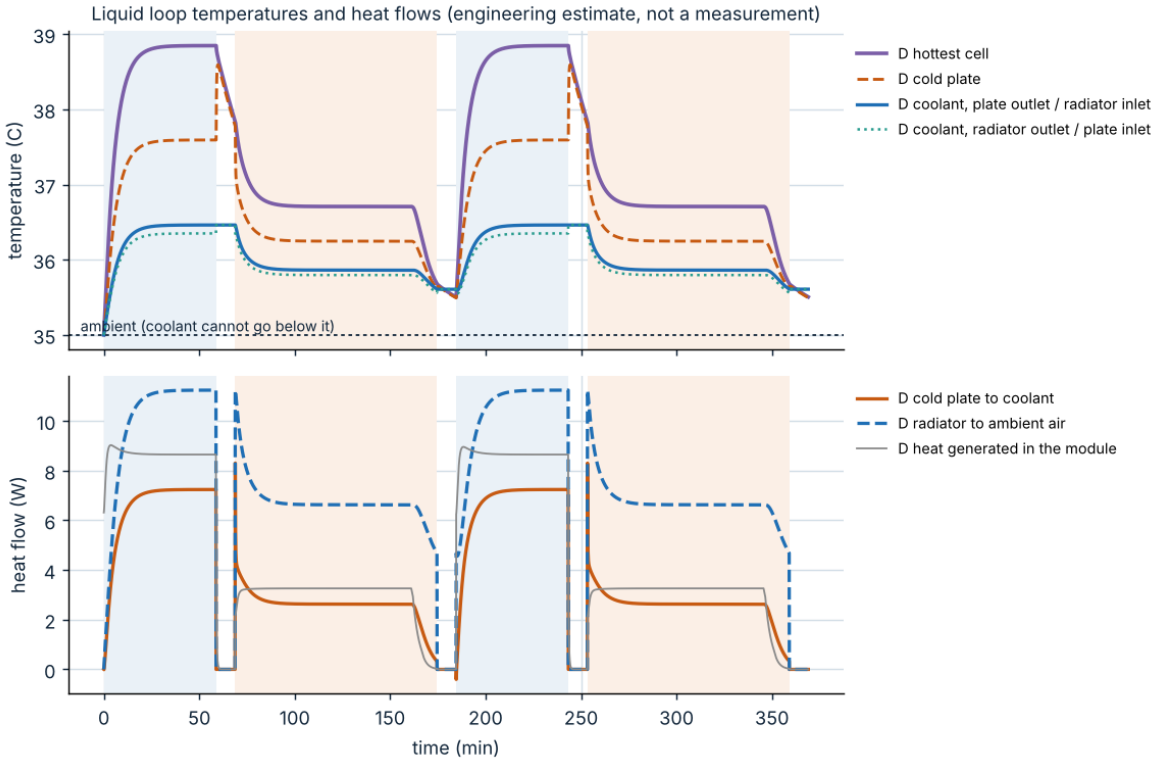


Figure 4: Design D: hottest cell, plate and coolant temperatures, and heat flows. The coolant cannot fall below ambient.

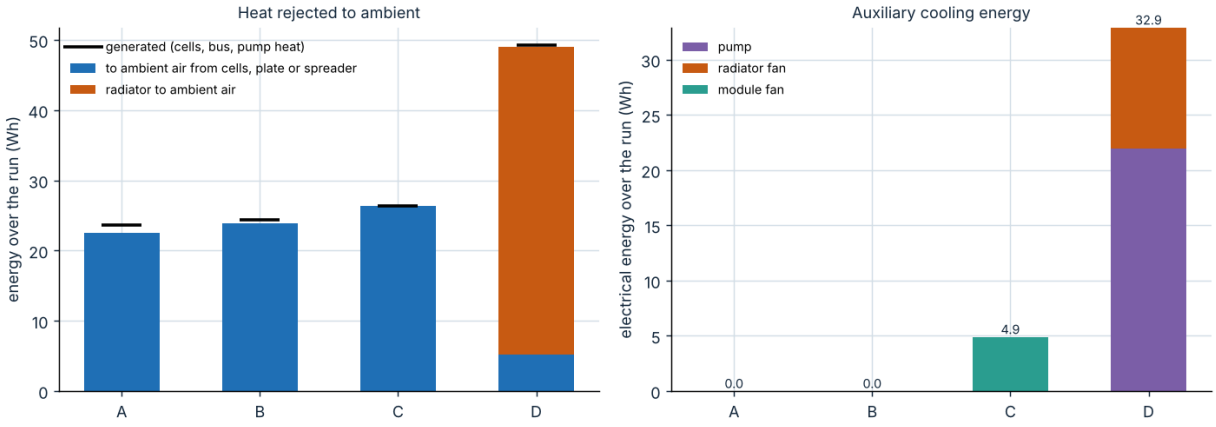


Figure 5: Where the heat goes (left) and what the cooling hardware costs in electrical energy (right), whole two-cycle run.

13.8 Ambient temperature, discharge rate and repeated cycles

Three cycles, 0.6C charge (limited to the discharge rate when that is lower). Peak cell temperature in °C; the text below gives the highest ambient at which the radiator can hold the coolant at the 45 °C requirement with the mean heat load of each case.

Ambient (°C)	Discharge	A	B	C	D
25	0.5C	34.0	29.7	27.3	26.5
25	1C	48.9	39.7	33.2	29.6
25	1.5C	65.7	51.0	40.6	34.0
35	0.5C	42.5	38.9	36.9	36.3
35	1C	57.4	47.7	41.8	38.8
35	1.5C	66.1	60.0	48.9	42.5
45	0.5C	51.5	48.4	46.6	46.2
45	1C	62.8	57.4	51.3	48.4
45	1.5C	63.6	60.3	58.8	52.1

Table 12: Peak cell temperature over three cycles (°C). Values above 45 are outside the requirement.

Within the 45 °C requirement: A 2, B 3, C 5, D 6 of 9 cases. At 25 °C and 35 °C ambient D is the coolest design at every rate, by 0.6 to 6.6 K below C. At 45 °C ambient no design stays within the requirement and none completes a charge: the BMS charge trip does not release until the cells fall below 40 °C, and the radiator cannot take the coolant below ambient. The radiator can hold the coolant at the requirement temperature only up to an ambient of 43.4 to 44.5 °C (mean heat load of each case), so liquid cooling with a radiator to ambient air does not remove the need for a cooler environment or a chiller above that. The peak of the last cycle never exceeds the peak of the first cycle by more than 0.2 K for D (no accumulation) in the cases below 45 °C.

13.9 Fault cases

Design D, 35 °C, 1C, the BMS temperature trips unchanged.

Fault	Peak (°C)	Mean (°C)	Min above 45	Result
Healthy loop	38.8	37.3	0	within the requirement
Flow reduced to 50 %	38.8	37.3	0	within the requirement
Flow reduced to 25 %	38.8	37.3	0	within the requirement
Radiator fan failed	41.7	39.3	0	within the requirement
Pump stops at 10 min	51.6	43.0	105	above the 45 C charge limit, BMS blocks charging
No flow from the start	51.6	43.2	111	above the 45 C charge limit, BMS blocks charging

Table 13: Coolant-side faults for design D.

Cutting the flow to 25 % changes the peak by less than 0.1 K, because the design flow is much larger than needed. A radiator-fan failure raises the peak to 41.7 °C. A pump failure is the dangerous case: after the pump stops at 10 minutes the peak reaches 51.6 °C, above the 45 °C charge limit, and the BMS blocks charging for about 58 min. The BMS sees only cell temperature, so it detects the fault late.

13.10 Safety and engineering risks outside the model

- **Electrical isolation.** The cans of 21700 cells sit at the potential of their series group, up to 16.8 V across the module. The pad must be a dielectric with a rated breakdown voltage, and the check LQ4 only records the design rule. An insulation-resistance test is needed.
- **Coolant leakage.** Water-glycol is conductive. A leak at the cells or busbars is a short-circuit and corrosion risk. Not modelled; use a sealed plate, keep fittings off the electrical side, add leak detection and pressure-test the loop.

- **Flow and pump failure.** The fault cases show the temperature consequence. The BMS has no flow sensor, so a pump fault is caught late, by the temperature trips.
- **Thermal interface resistance.** It is lumped in the pad conductivity and thickness. Contact pressure and ageing are not modelled.
- **Over-temperature protection.** The 45 °C charge and 60 °C discharge trips act on the sensor cell. In design D the spread is below 0.2 K, so the corner sensor reads the module well.
- Pressure drop, flow distribution, air locks, fouling, freezing and boiling are not modelled.

13.11 Limits of the liquid model

- Two thermal nodes for the loop and effectiveness conductances: no flow network, no pressure drop, no transport delay. The default time constant of the coolant node ($C_c/G_r \approx 162\,\mathrm{s}$) is short against the 58 min discharge, so a delay would not change the peak much, but it is not tested.
- All parameters in the table above are estimates or placeholders. Mass, volume and cost of the loop hardware are stand-ins, not quotes. The result that D is heavy and expensive follows from those placeholders and has to be re-run with real parts.
- The loop has no refrigeration. At or above the 45 °C requirement it cannot cool the cells to the requirement.
- The flow is oversized for about 9 W of heat (coolant rise 0.08 K). The peak barely moves from 30 down to 3 g/s, and rises only below about 1 g/s. A smaller pump is a possible saving that this model does not cost.

14 Ranking and recommendation

Source: `rank` *in* `bms/checks.py`. For each indicator j with weight w_j and value $x_{j,d}$ for design d, the value is min–max normalised across the designs:

$$\hat{x}_{j,d} = \frac{x_{j,d} - \min_d x_{j,d}}{\max_d x_{j,d} - \min_d x_{j,d}}, \qquad u_{j,d} = \begin{cases} \hat{x}_{j,d} & \text{higher is better (usable energy)} \\ 1 - \hat{x}_{j,d} & \text{lower is better (all others)} \end{cases} \tag{32}$$

$$S_d = \sum_j w_j \, u_{j,d}, \qquad \sum_j w_j = 1. \tag{33}$$

If all designs have the same value for an indicator, its contribution is 0 for all. Designs are then ordered by (1) number of datasheet fails, (2) number of electrochemical fails, (3) number of requirement fails, (4) number of liquid-loop fails, (5) score S_d in descending order. A design is *acceptable* if it has no datasheet, electrochemical, requirement or liquid-loop fail. The ranking is computed from the simulated results of every design in the comparison, so it does not assume liquid cooling wins or loses. The first row is the recommendation. Default weights: temperature 0.30, energy 0.15, charge time 0.15, mass 0.15, cost 0.15, volume 0.10. The sidebar sliders are rescaled to sum to 1.

Indicator	Weight	u (A)	u (B)	u (C)
Peak temperature	0.30	0.000	0.618	1.000
Usable energy	0.15	0.000	0.793	1.000
Charge time	0.15	0.000	1.000	0.952
Mass	0.15	1.000	0.000	0.837
Cost	0.15	1.000	0.000	0.584
Volume	0.10	1.000	0.688	0.000
Weighted score	1.00	0.400	0.523	0.806

Table 14: Worked ranking example. Design C scores highest (0.806) and has no datasheet or requirement fail, so it is the recommendation. Design A scores lowest (0.400) and also has a datasheet fail, so it is ranked last by every criterion.

15 Graphs and tables: what is plotted

Source: `bms/plots.py`, `run_report.py`, `app.py`.

15.1 Hottest cell temperature over the duty cycle

Plotted: $x = t/60$ in minutes, $y = T_{\max}(t)$ for each design. Blue shading marks discharge phases and orange marks charge phases (taken from the first design, since all designs run the same schedule). The dashed line is the 45 °C requirement, which is also the datasheet charge limit, so one line carries both labels. The legend value is T_{peak}. In design A the curve flattens near 45 °C during charge: the corner sensor holds the BMS at its trip level while the hottest cell sits several kelvin above.

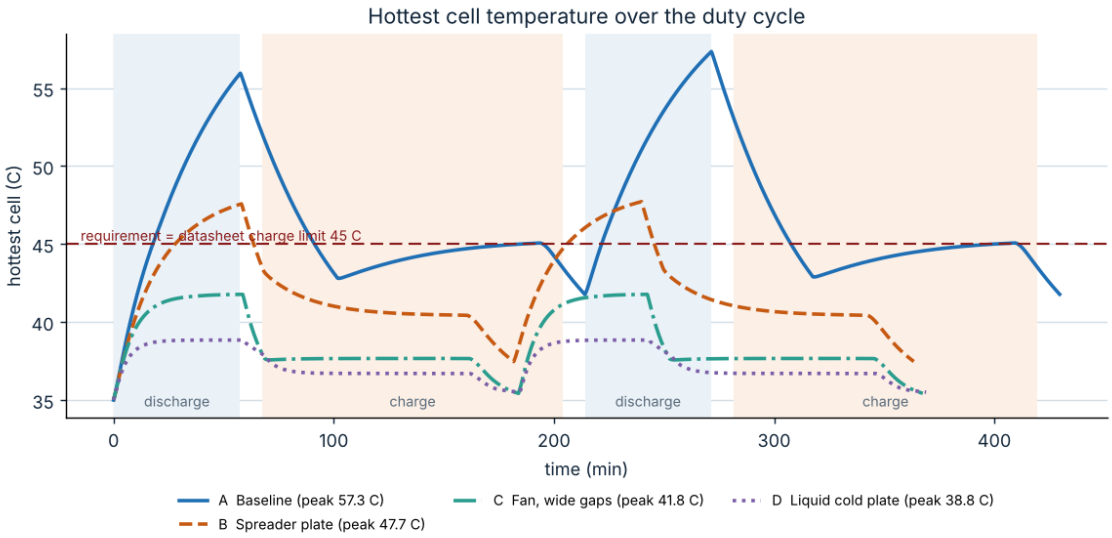


Figure 6: Hottest cell temperature, default scenario.

15.2 Indicators against requirements

Six bar panels: T_{peak}, E_{use}, t_{chg}, mass, cost and volume. The dashed line in each panel is the requirement from Section 11. In the temperature panel the vertical axis starts at $\min(\text{values}, \text{limit}) - 8$ so that differences are visible.

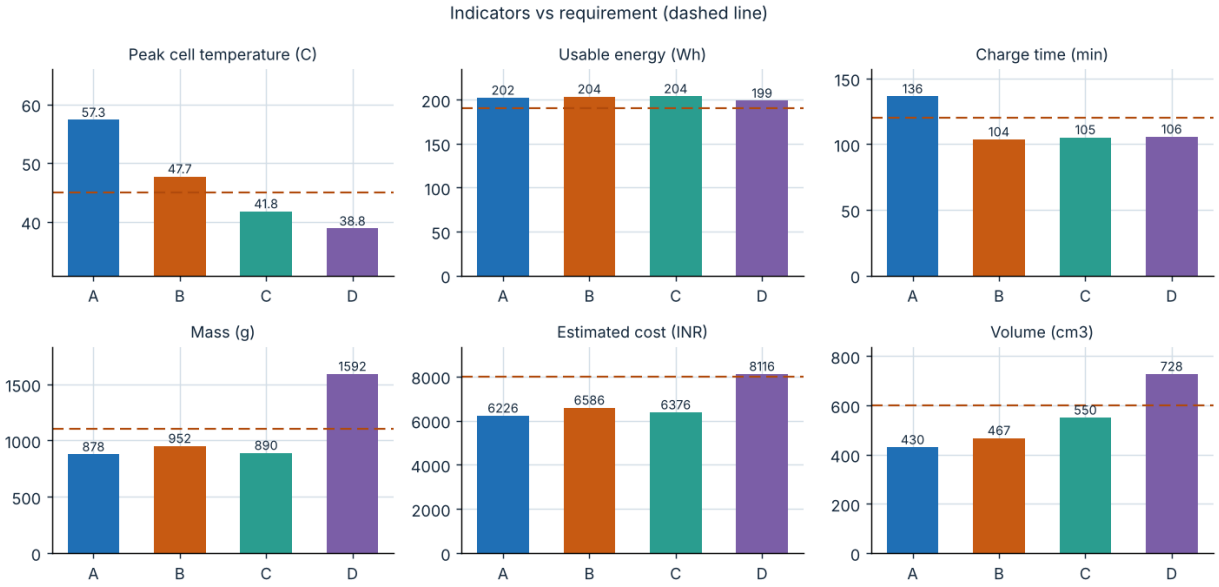


Figure 7: Indicators against requirements.

15.3 Cell temperature map

For each design, $t^* = \arg\max_t T_{\max}(t)$. The map shows the grid $G[i, g] = T_{(g-1)n_{\text{P}}+i}(t^*)$ with series groups S1 to S4 as columns and parallel positions P1 to P3 as rows. All maps share one colour scale from the lowest to the highest cell temperature over all designs and all times, so colours compare across designs. In A the interior cells (S2 and S3, row P2) are the hottest. In B and C the plate or the airflow evens out the temperature.

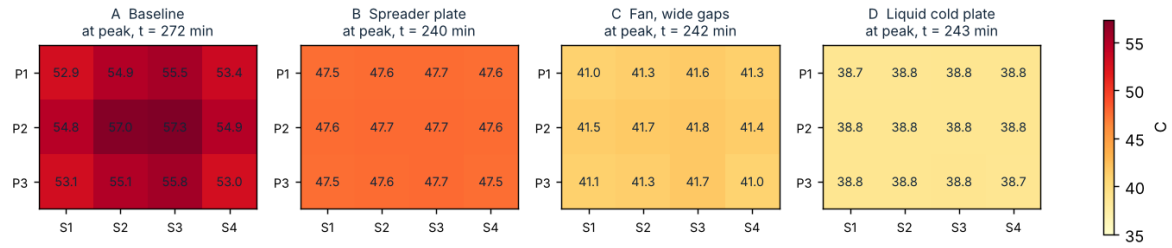


Figure 8: Cell temperature at the moment of the peak.

15.4 Electrical behaviour

Three panels against time: V_{mod}, I (positive for discharge, negative for charge), and SOC as $100\,s$. The voltage steps up when the current reverses because the IR_0 drop changes sign, then creeps as V_1 relaxes during rest, and the current falls during the CV tail.

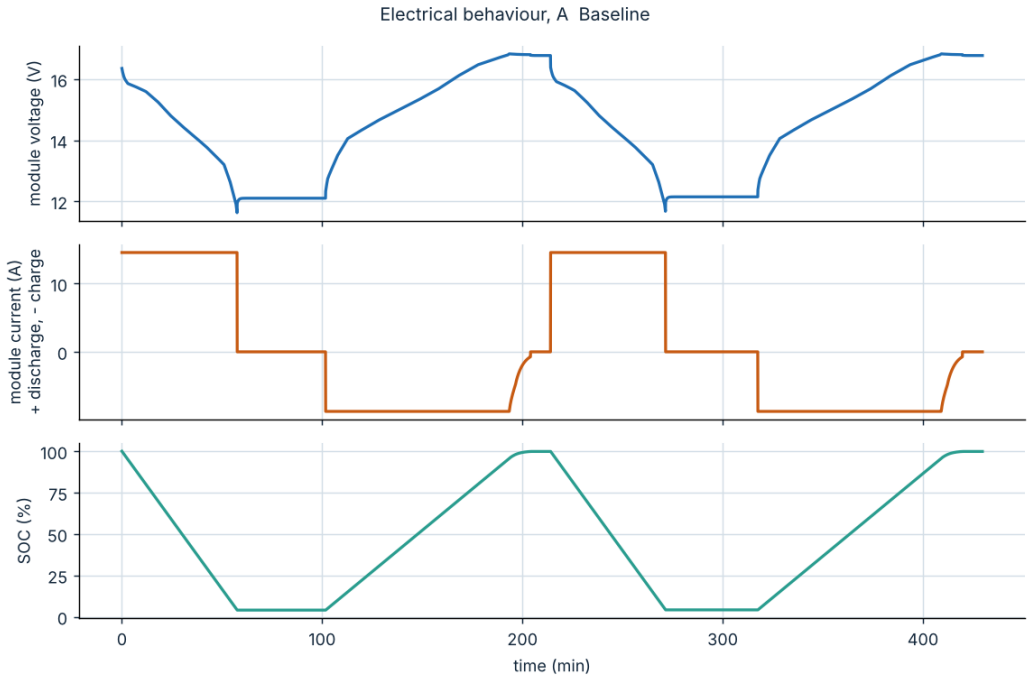


Figure 9: Voltage, current and SOC for design A.

15.5 Sweeps

For each value x the model runs one discharge and charge cycle ($\Delta t = 10\,\mathrm{s}$) and records T_{peak}. In the discharge-rate sweep $c_{\mathrm{dis}} = x$ and $c_{\mathrm{chg}} = \min(x, 0.6)$, from 0.5C to 1.5C, the datasheet discharge maximum. In the ambient sweep $T_{\mathrm{amb}} = x$ from 15 to 45 °C. The largest rate that stays at or below 45 °C on a 0.05C grid, one cycle, 35 °C ambient, is 0.55C for A, 0.85C for B and 1.25C for C.

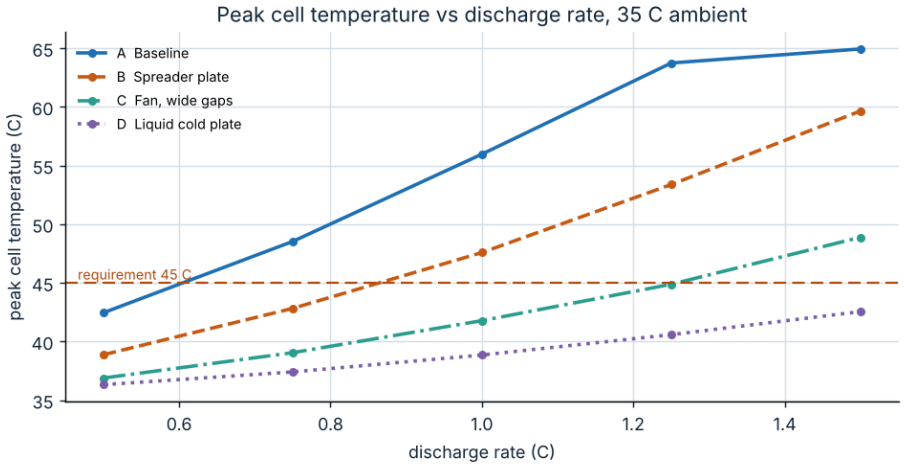


Figure 10: Peak cell temperature versus discharge rate.

15.6 Tables in the dashboard

- **Ranking**: Section 14.
- **Indicators**: Section 9, round-trip efficiency shown in %.
- **Checks**: Section 11, with value, limit, unit and reason.

- **Traceability**: one row per requirement with component, assumption, model, result (value against limit, PASS or FAIL) and the validation or source. Downloadable as CSV.
- **Sensitivity**: Section 17.

16 Validation

Source: `bms/validate.py`, `tools/pybamm_crosscheck.py`. All tests are isothermal, with one ideal cell (no spread, no heating). Because the specification has no rate or charge-time data, each row is labelled *independent* (the model was not set to match it) or *input reproduced* (it only confirms that an input was entered correctly).

16.1 Against the datasheet and the chart points

Test	Condition	Datasheet or chart	Model	Diff.	Note
Energy at 0.2C, 25 C	0.97 A to 2.5 V	17.6 Wh min, 18.2 Wh nominal	17.8 Wh	-0.4 Wh vs nominal	independent: OCV from the electrochemical model, R from DS
Mean discharge voltage at 0.2C	0.97 A to 2.5 V	3.63 V	3.68 V	+0.05 V	independent
DC resistance, 0.5C pulse	30 s, 25 C, 50 % SOC	30 +/- 6 mOhm	31.4 mOhm	+1.4 mOhm	input reproduced, not independent
DC resistance, 0.5C pulse	0.5 s, 25 C, 50 % SOC	21 to 25 mOhm (3 cells)	24.9 mOhm	top of range	independent: R0 is fitted to the electrochemical model, not to these points
Resistance ratio to 25 C	10 C	2.96 (chart means)	2.96	+0.00	input reproduced, not independent. Chart read by eye
Resistance ratio to 25 C	45 C	0.61 (chart means)	0.61	+0.00	input reproduced, not independent. Chart read by eye
Energy vs 25 C, resistance only	0 C, 0.2C	80 %	94 %	+14 points	independent: resistance alone does not explain the loss, so the datasheet table is added
Energy vs 25 C, with DS table	0 C, 0.2C	80 %	79 %	-1 points	input reproduced, not independent
Energy vs 25 C, resistance only	-10 C, 0.2C	70 %	86 %	+16 points	independent: resistance alone does not explain the loss, so the datasheet table is added
Energy vs 25 C, with DS table	-10 C, 0.2C	70 %	65 %	-5 points	input reproduced, not independent
Energy vs 25 C, resistance only	60 C, 0.2C	95 %	100 %	+5 points	independent: resistance alone does not explain the loss, so the datasheet table is added
Energy vs 25 C, with DS table	60 C, 0.2C	95 %	97 %	+2 points	input reproduced, not independent

Table 15: Model against datasheet (DS) and chart (BD).

Reading. The OCV curve and $R_0 + R_1$ reproduce the datasheet energy at 0.2C: 17.8 Wh against 17.6 minimum and 18.2 nominal, with a mean voltage 0.05 V above the nominal 3.63 V. The 0.5 s resistance of 24.9 mΩ is at the top of the 21 to 25 mΩ points on the BD chart and was not fitted. Resistance alone predicts 94 % energy at 0 °C and 86 % at −10 °C against 80 and 70 % in the datasheet, which is why the datasheet table is added for temperature; those rows then match by construction. At 60 °C the datasheet loses 5 % and the resistance model loses none.

16.2 Against the electrochemical model

The PB set is run in the DFN model at 25 °C by `tools/pybamm_crosscheck.py`. The set has no temperature dependence (activation energies and entropic coefficients are zero), so it cannot check cold or hot behaviour. Charge times are compared in units of C-hours (time multiplied by the C-rate) so that the 5.12 Ah DFN cell and the 4.85 Ah rated cell are on one basis. The last column of the table runs the circuit with the DFN's own resistance, which separates model structure from the resistance difference.

Test	Condition	DFN	Circuit	Diff.	Circuit, DFN R	Type
DC resistance, 0.5C, 30 s, 50 % SOC	25 C	38.4 mOhm	31.4 mOhm	-7.0 mOhm	-	DFN is above the datasheet 30 +/- 6 mOhm; circuit is set to the datasheet
DC resistance, 0.5C, 0.5 s, 50 % SOC	25 C	27.1 mOhm	24.9 mOhm	-2.2 mOhm	-	measured cells: 21 to 25 mOhm (BD chart)
Capacity vs 0.2C	0.2C, 25 C	100.0 %	100.0 %	+0.0	100.0 %	independent
Energy vs 0.2C	0.2C, 25 C	100.0 %	100.0 %	+0.0	100.0 %	independent
Capacity vs 0.2C	0.5C, 25 C	99.1 %	99.7 %	+0.6	99.6 %	independent
Energy vs 0.2C	0.5C, 25 C	97.2 %	98.3 %	+1.1	97.9 %	independent
Capacity vs 0.2C	1.0C, 25 C	97.5 %	99.2 %	+1.7	99.0 %	independent
Energy vs 0.2C	1.0C, 25 C	93.1 %	95.4 %	+2.4	94.4 %	independent
Capacity vs 0.2C	1.5C, 25 C	95.8 %	98.7 %	+2.9	98.3 %	independent
Energy vs 0.2C	1.5C, 25 C	88.9 %	92.6 %	+3.7	90.9 %	independent
Charge time to end current, in C-hours	0.3C CC-CV, 25 C	1.221 h x C	1.056 h x C	-14 %	1.075 h x C (-12 %)	independent, 50 mA end current
Charge time to end current, in C-hours	0.5C CC-CV, 25 C	1.200 h x C	1.076 h x C	-10 %	1.101 h x C (-8 %)	independent
Charge time to end current, in C-hours	0.6C CC-CV, 25 C	1.277 h x C	1.110 h x C	-13 %	1.149 h x C (-10 %)	independent
Charge time to end current, in C-hours	0.7C CC-CV, 25 C	1.364 h x C	1.150 h x C	-16 %	1.204 h x C (-12 %)	independent, C/20 end current

Table 16: Equivalent circuit against the PyBaMM DFN model, 25 °C.

Reading. With the datasheet resistance the circuit gives energy within 2.4 points of the DFN at 1C and 3.7 at 1.5C, and these are optimistic because the DFN cell is 28 % more resistive (38 against 30 mΩ). With the DFN's resistance the gap is smaller. Charge time is 10 to 16 % shorter than the DFN at 0.3C to 0.7C, so the charge times in this report are lower bounds. A second polarisation branch did not improve the pulse fit, so the remaining difference is not a matter of branch count. A plausible cause is the SOC dependence of diffusion in the DFN, which a fixed linear branch does not follow; this was not tested.

17 Sensitivity to the uncertain inputs

Source: `bms/sensitivity.py`. One input at a time is moved between the two levels shown, with the default scenario and $\Delta t = 10$ s, and the four designs are re-ranked. The last four inputs apply to the liquid loop only, so A, B and C do not move.

Input	Level	A (°C)	B (°C)	C (°C)	D (°C)	Order
Base case	-	57.4	47.7	41.8	38.8	C > B > D > A
Natural convection h, +/- 30 %	7	61.1	51.6	42.5	39.0	C > D > B > A
Natural convection h, +/- 30 %	13 W/m2K	54.0	45.3	41.2	38.7	C > B > D > A
Gap exposure length, 2 / 4.5 mm	2	54.3	46.8	41.1	38.8	C > B > D > A
Gap exposure length, 2 / 4.5 mm	4.5 mm	60.7	48.6	43.2	38.9	C > B > D > A
Forced convection coefficient, +/- 30 %	-30 %	57.4	47.7	43.1	38.8	C > B > D > A
Forced convection coefficient, +/- 30 %	+30 %	57.4	47.7	40.8	38.8	C > B > D > A
Cell specific heat, 800 / 1100	800	58.2	47.9	41.8	38.8	C > B > D > A
Cell specific heat, 800 / 1100	1100 J/kgK	55.9	47.3	41.8	38.8	C > B > D > A
Cell resistance, +/- 20 % (DS tolerance)	-20 %	53.6	45.6	40.7	38.3	C > B > D > A
Cell resistance, +/- 20 % (DS tolerance)	+20 %	61.1	49.9	42.8	39.4	C > B > D > A
Resistance spread, 0 / 10 %	0 %	57.7	47.8	42.0	38.8	C > B > D > A
Resistance spread, 0 / 10 %	10 %	57.0	47.6	42.0	38.9	C > B > D > A
Ambient temperature, 30 / 40 C	30 C	53.0	43.5	37.5	34.2	C > B > D > A
Ambient temperature, 30 / 40 C	40 C	60.6	52.4	46.3	43.5	C > D > B > A
Thermal pad conductivity, 1.0 / 2.5	1.0	57.4	47.8	41.8	39.3	C > B > D > A
Thermal pad conductivity, 1.0 / 2.5	2.5 W/mK	57.4	47.6	41.8	38.4	C > B > D > A
Radiator UA, +/- 30 %	-30 %	57.4	47.7	41.8	39.3	C > B > D > A
Radiator UA, +/- 30 %	+30 %	57.4	47.7	41.8	38.6	C > B > D > A
Plate-to-coolant UA, +/- 30 %	-30 %	57.4	47.7	41.8	39.2	C > B > D > A
Plate-to-coolant UA, +/- 30 %	+30 %	57.4	47.7	41.8	38.6	C > B > D > A
Coolant flow, 50 / 150 %	50 %	57.4	47.7	41.8	38.8	C > B > D > A
Coolant flow, 50 / 150 %	150 %	57.4	47.7	41.8	38.9	C > B > D > A
Liquid loop hardware cost, 50 / 200 %	50 %	57.4	47.7	41.8	38.8	C > B > D > A
Liquid loop hardware cost, 50 / 200 %	200 %	57.4	47.7	41.8	38.8	C > B > D > A

Table 17: Peak cell temperature of each design and the resulting order.

Design C ranks first in 25 of 25 cases. D ranks second in 2 and third in 23; B ranks second in 23. A is last in every case. D has the lowest peak temperature in all 25 cases, but it stays behind C (and usually B) because it fails the mass, volume and cost requirements, which do not depend on these inputs. Radiator UA and plate-to-coolant UA of ±30 % move the peak of D by -0.2 to +0.5 K and -0.2 to +0.4 K, and the coolant flow (50 to 150 %) by +0.0 to +0.1 K, so the temperature result of D is not very sensitive to those estimates around the default point; the cost placeholder does not change the order. The absolute temperatures of A move by about ±3.5 K for the natural convection coefficient and ±3.8 K for the cell resistance, and those of C by about ±4.5 K for a 5 K change of ambient. The 45 °C requirement is met by C in 24 of 25 cases (not at 40 °C ambient, 46.3 °C), by

D in 25 of 25 (highest 43.5 °C, at 40 °C ambient: 43.5), by B in 1 and by A in none.

18 Automated tests

`tests/test_model.py` (run with `python tests/test_model.py` or `pytest`) checks: energy balance over a full run (terminal energy equals OCV energy minus heat minus change of stored polarisation energy, within 3 %); SOC and voltage bounds; repeatability with a fixed seed and sensitivity to the seed; temperature symmetry without spread; the datasheet anchors ($0.2Q = 0.97$ A, $0.7Q = 3.395$ A, $1.5Q = 7.275$ A, DCIR inside 30 ± 6 mΩ, 0.2C energy inside 17.6 to 18.4 Wh); the current-limit functions; monotonic resistance against temperature; charge-time trends; the ranking order; that no charge current flows while a cell is below 0 °C; the plating limit falls with SOC, starts at the 1.5C cap and sits between 0.6C and 0.8C at 90 % SOC; check E1 passes at 0.6C and fails at 0.8C; and that cells below 25 °C are left out of the plating ratio and the time is reported instead (14 tests). Design D adds 15 more: A, B and C unchanged against recorded reference values; heat balance and coolant balance; stability of the network with the pump running, weak and stopped; step-size independence; coolant never below ambient; every liquid parameter moves the temperature in the right direction; pump failure and zero flow; BMS trip on a pump failure; auxiliary energy counted in the net energy; checks LQ1 to LQ4 only for design D; the radiator cannot cool below ambient; the ranking follows the simulated results and does not declare D the winner by default; no drift over three cycles (29 tests in total).

19 Assumptions and limitations

- The M50 specification is a 2016 tentative draft with no OCV, rate, charge-time or thermal data. The OCV curve is from a research parameter set (PB), and the resistance ratios against temperature are read by eye from a chart (BD).
- Lumped thermal model: one temperature per cell, no gradients inside a cell, no CFD.
- One polarisation branch fitted at 25 °C. Charge time is 10 to 16 % shorter than the electrochemical model and energy at 1C is 2 to 3 points higher, so both are optimistic. The branch time constant follows R_1 with temperature, which is an assumption.
- The electrochemical layer is a 25 °C screen (Section 12). It is not used for cold behaviour, temperature, energy or charge time, and it does not measure plated lithium.
- Liquid cooling (Section 13) is a lumped engineering estimate with no measurement behind it: coolant properties, flow, plate and radiator conductances, pump and fan power, pad properties and the hardware mass, volume and cost are all estimates or placeholders.
- No ageing. The datasheet states 500 cycles to at least 80 % of minimum energy under a 0.33C cycle (DS 4.3.4), and BD shows life falling quickly above 1C charge. Neither is tracked.
- Cold and hot energy loss comes from the DS 4.4.4 table, measured at 0.2C. Resistance below 10 °C is an extension of the chart slope.
- Reversible heat is not modelled.
- The convection coefficients, gap exposure scale s_0, forced-air law, cell specific heat and thermal pad properties are engineering estimates and need a bench test. Section 17 shows their effect.
- Heat conduction through nickel strips and cell ends is ignored, so hotspots are slightly pessimistic.
- Cost and non-cell mass are placeholders. The cell price needs a supplier quote.

- The requirement values are editable defaults, not figures from the problem statement.
- Simulation only. It is not a safety certification of a real pack.

A Parameter table with sources

Parameter	Value	Unit	Source
Capacity (rated minimum, defines 1C)	4.85	Ah	DS 2.1, 2.9
Nominal voltage	3.63	V	DS 2.2
Charge voltage	4.2	V	DS 2.5
Discharge cut-off	2.5	V	DS 2.7
Cell mass	0.068	kg	DS 2.10
Diameter (max)	0.0211	m	DS 3.2
Height (max)	0.07015	m	DS 3.2
Max continuous discharge, 5 to 60 C (1.5C)	7.275	A	DS 2.9
Max continuous discharge, -20 to 5 C (0.3C)	1.455	A	DS 2.9
Standard charge current (0.3C)	1.455	A	DS 2.4
Max charge current, 0 to 25 C (0.3C)	1.455	A	DS 2.6
Max charge current, 25 to 50 C (0.7C)	3.395	A	DS 2.6
Charge temperature, min	0	C	DS 5.1
Charge temperature, max	45	C	DS 5.1
Charge release below	40	C	ASSUMPTION: not in datasheet, 5 K hysteresis
Discharge temperature, min	-20	C	DS 5.1
Discharge temperature, max	60	C	DS 5.1
Discharge release below	55	C	ASSUMPTION: not in datasheet, 5 K hysteresis
Charge end current, standard charge	0.05	A	DS 2.4
Charge end current, faster charge (C/20)	0.05	C	DS 4.2.3
Impedance at 1 kHz (max)	25	mOhm	DS 4.3.1
DC resistance, 25 C, 50 % SOC, 30 s (anchor)	30	mOhm	DS 4.3.2 (30 +/- 6)
Series resistance R0, 25 C	24.78	mOhm	FIT to PB, scaled so R0 + R1 (1 - e^(-30/tau)) = 30
Polarisation resistance R1, 25 C	14.07	mOhm	FIT to PB, same scaling

Parameter	Value	Unit	Source
Polarisation time constant, 25 C	65	s	FIT to PB (tools/fit_polarisation.py)
Specific heat	900	J/kg/K	ASSUMPTION: typical 800 to 1100, no M50 value in sources
Cell price	450	INR	ASSUMPTION: placeholder, use a real quote
Open-circuit voltage curve	16 points	V vs SOC	PB: Chen2020 single-particle model at C/50, 25 C (not a datasheet curve)
Resistance and tau vs temperature	x2.96 at 10 C, x1 at 25 C, x0.61 at 45 C	-	BD: DCIR chart read by eye (+/- 2 mOhm). Below 10 C extended, above 45 C held
Energy retention vs temperature	70 % at -10 C, 80 % at 0 C, 100 % at 25 C, 95 % at 60 C	-	DS 4.4.4 (0.2C discharge)

Design parameters (defaults).

Parameter	A Baseline	B Spreader	C Fan, wide gaps
Layout	4S3P	4S3P	4S3P
Gap between cells	1.5 mm	1.5 mm	4.0 mm
Natural convection h_{nat}	10 W/m^2/K	10 W/m^2/K	10 W/m^2/K
Fan air speed / power	none	none	2 m/s / 0.8 W
Spreader plate	none	125 cm^2, 2 mm Al	none
Thermal pad	–	1.5 W/m/K, 1 mm, 5 cm^2 per cell	–
BMS sensor on	corner cell	corner cell	corner cell
Interconnect resistance	4 mΩ	4 mΩ	4 mΩ

Scenario and requirement defaults. Ambient 35 °C; discharge 1C; charge 0.6C; 2 cycles; rest 10 min; resistance spread 5 %; seed 7. Requirements: $T_{\text{peak}} \leq 45$ °C; $E_{\text{use}} \geq 190$ Wh; $t_{\text{chg}} \leq 120$ min; mass ≤ 1100 g; cost $\leq$ INR 8000; volume ≤ 600 cm^3; load window 10 to 16.8 V; cell gap ≥ 1 mm.

B Where each formula lives in the code

Item	Location
OCV, $f(T)$, energy retention, limits, termination current	`bms/cell.py`: `ocv`, `r_temp_factor`, `cap_retention`, `i_chg_limit`, `i_dis_limit`, `i_term`
Cell data and sources	`bms/cell.py`: `Cell`, `cell_param_table`
Thermal matrices	`bms/sim.py`: `build_thermal`, `h_effective`
Cycle loop, protection, heat	`bms/sim.py`: `simulate`
Liquid loop: conductances, matrix, indicators	`bms/sim.py`: `liquid_parts`, `build_thermal`, `liquid_indicators`
Liquid parameters, mass, cost, volume	`bms/design.py`: `Design`, `LIQUID_SOURCES`, `mass_g`, `cost_inr`, `volume_cm3`
Ambient and rate study, fault cases	`bms/study.py`: `ambient_rate_study`, `failure_modes`
Indicators	`bms/sim.py`: `indicators`
Mass, cost, volume	`bms/design.py`: `mass_g`, `cost_inr`, `volume_cm3`
Checks, traceability, ranking	`bms/checks.py`: `run_checks`, `traceability`, `rank`
Validation	`bms/validate.py`: `validation_table`
Sensitivity	`bms/sensitivity.py`: `run`
Plating limit and live replay	`bms/echem.py`: `plating_limit_c`, `plating_limit_a`, `replay`
OCV table, polarisation fit, DFN cross-check, plating map, replay (need PyBaMM)	`tools/make_ocv.py`, `tools/fit_polarisation.py`, `tools/pybamm_crosscheck.py`, `tools/make_plating_map.py`, `tools/echem_replay.py`
Tests	`tests/test_model.py`
Figures	`bms/plots.py`
Report and dashboard	`run_report.py`, `app.py`